



**FACULTY OF COMPUTING AND INFORMATION
TECHNOLOGY**

BMMS2074 Statistics for Data Science

Assignment

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Tutorial Group: 4

Title : Social Media Popularity for Facebook between year 2009 to 2023

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Date of submission : 25 September 2024

PART-II: Depth of Knowledge Assessment Rubrics

Program Learning Outcomes	Evaluation Criteria	Competency Levels				Score
		0-3 Unsatisfactory	4-7 Fair	8-11 Good	12-15 Outstanding	
Critical Thinking and Problem Solving (75%)	Written Communication (15%)	Attempts to use a consistent system for basic organization; minimal attempts to use sources to support ideas in the writing and these sources may not be correctly documented using an appropriate referencing style and/or may not be fully relevant to the task at hand.	Follows expectations appropriate to a specific discipline and/or writing task for basic organization, and content; use credible and/or relevant sources to support ideas and to document these sources properly using APA or Harvard referencing style.	Demonstrates consistent use of important conventions particular to a specific discipline and/or writing task; consistently use credible, relevant sources appropriate to the discipline and genre to support ideas and documents sources with few errors or exceptions using APA or Harvard referencing style.	Demonstrates detailed attention to and successful execution of a wide range of conventions particular to a specific discipline and/or writing task (including organization, content, formatting, and stylistic choices); synthesize a range of high-quality, credible, relevant sources that are appropriate for the discipline and genre to develop ideas and fully documents these sources using APA or Harvard referencing style.	
	Problem Solving Strategy and Approaches (15%)	Unable to identify an approach to possible solution.	Identifies a possible but very general approach to a solution without a clear sense of the steps to solve the problem.	Identifies a reasonable and problem specific possible approach to a solution with some sense of steps to be undertaken to reach a solution.	Identifies at least two reasonable and problem specific possible approaches to a solution. Outlines several steps in detail and/or identifies another reasonable and problem specific possible approach.	
	Analysis (15%)	Demonstrates emerging understanding of the data analysis without showing evident e.g. diagrams, models, timelines, illustrations, explanations, or a series of steps in the report. The report fails to tie into basic concepts	Demonstrates moderate understanding of the data analysis that are somewhat evident e.g. diagrams, models, timelines, illustrations, explanations, or a series of steps within the report. The report may fail to tie into basic concepts and	Demonstrates considerable understanding of the data analysis and are evident e.g. diagrams, models, timelines, illustrations, explanations, or a series of steps within the report. The report ties into basic concepts and builds on prior	Demonstrates in-depth/ thorough understanding of the data analysis and are clearly evident e.g. diagrams, models, timelines, illustrations, explanations, or a series of steps throughout the report. The report ties into basic concepts and builds on prior knowledge.	

		and build on prior knowledge.	build on prior knowledge.	knowledge.		
	Critical Thinking and Perspective-Taking (15%)	Specific position is stated but is simplistic and obvious.	Information is presented with some interpretation or evaluation, but not enough to develop a coherent analysis or synthesis.	Specific position takes into account the complexities of an issue and acknowledges other viewpoints.	Questions are examined from a range of viewpoints, taking into account the complexities of an issue.	
	Conclusions and Related Outcomes (Implications and Consequences) (15%)	Conclusion is inconsistently tied to some of the information discussed; related outcomes (consequences and implications) are oversimplified.	Conclusion is logically tied to information (because information is chosen to fit the desired conclusion); some related outcomes (consequences and implications) are identified clearly.	Conclusion is logically tied to a range of information, including opposing viewpoints; related outcomes (consequences and implications) are identified clearly.	Conclusions and related outcomes (consequences and implications) are logical and reflect student's informed evaluation and ability to place evidence and perspective discussed in priority order.	
Total:						

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1.0 Introduction

The dataset "Social Media Popularity (2009-2023)" provides insights into social media engagement trends over a significant period, capturing the rise of various platforms and the evolving nature of digital communication. As social media has become a crucial element in marketing strategies and personal interactions, understanding the factors that drive user engagement is essential for businesses, researchers, and marketers. This dataset spans over a decade, reflecting changes in user behavior, platform popularity, and the impact of major global events (e.g., the COVID-19 pandemic) on social media usage.

The dataset is sourced from Kaggle, a popular online platform that provides data scientists and machine learning practitioners with tools and resources to build models, collaborate and also compete. It contains historical data on social media metrics collected from various platforms, including Facebook, Twitter, Instagram, and others. The data is organized on a monthly frequency, allowing for a granular analysis of trends and patterns over time.

2.0 Data Description

The zip folder that we have downloaded from Kaggle consists of two dataset and we have used the dataset which is named "social_media". The dataset has a total of 178 observations and 10 variables, such as the date of the measurement period with the format (YYYY-MM) starting from March 2009 to December 2023, the rest of the variables contains the percentage of share in the social media market, given as a percentage, rounded to 2 decimal places. If the share is less than 0.5%, the value 0 remains. These variables are Facebook, Pinterest, Twitter, StumbleUpon, Youtube, Instagram, Tumblr, Reddit and VKontakte. The variables that we will be focusing on are the date of the measurement period and the social media platform Facebook.

A consistent and complete dataset is crucial in order to generate analysis or do accurate prediction. However, we have found outlier data that has a date of March 2009 and has zero values for all percentages of the share of every social media variable. As the observation mentioned above does not provide any pattern or insights to our predictions, therefore we have decided to remove it to prevent outliers. After that, the entire dataset has no missing values, inconsistencies and anomalies, which means that it is ready to be used for analysis.

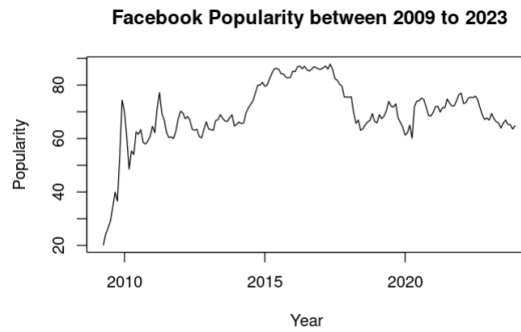


Figure 1 Monthly Facebook Popularity between 2009 to 2023

The above figure is the time series plot labeled “Social Media Popularity between 2009 to 2023”. The X-axis represents the time period from 2009 to 2023 and it tracks the progression of social media, Facebook’s share over the years. The Y-axis represents the percentage of the amount of shares, ranging from 20 to 80. From 2009 to 2012, there was a rapid increase in the share of social media , Facebook in the early years. This is possible due to the rise of social media platforms during the years along with the introduction of new smartphones. From 2013 to 2020, the shares stayed relatively stable but high with only a slight fluctuation. During this period, it could indicate the maturity of the social media platform, Facebook and its widespread adoption. From 2020 to 2023, a small decrease in share can be seen near the end of the graph. This occurrence could be due to user saturation, competition of other new social media platforms or changes in user behavior over time.

3.0 Exploratory Data Analysis (EDA)

3.1 Train Test Split

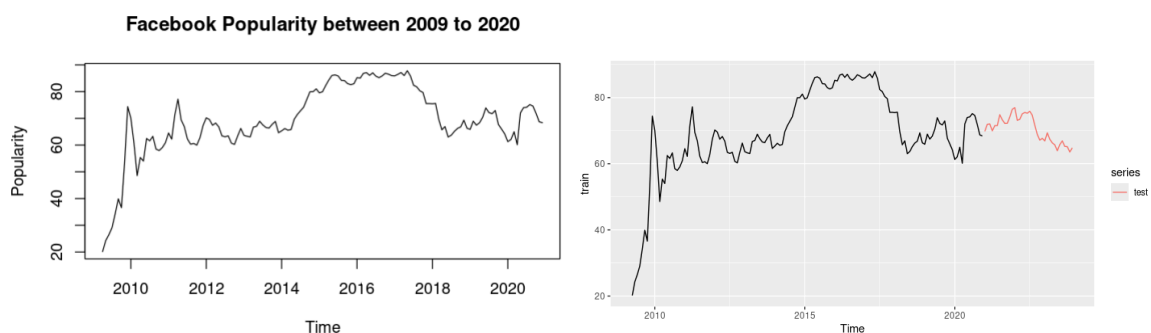


Figure 2 Monthly Facebook Popularity between year of 2009 to 2020 in Malaysia (Train Test Split)

We split the time series dataset by using an 80/20 train test split. 80% of the dataset will be used for modeling and 20% for validation purposes. Based on the entire given time span from 2009 to 2023 that we have, we set the data starting from 2009 to 2020 as our training data while 2021 to 2023 as our testing data. This is to ensure that the model created by using 80% of the dataset can also generalize well to new inputs by referring to the remaining 20% of the dataset.

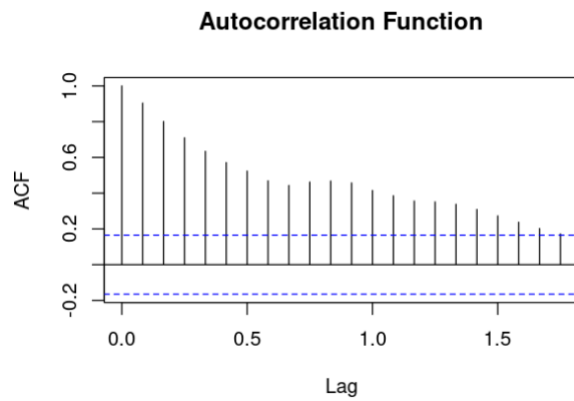


Figure 3 Correlogram of Facebook Popularity

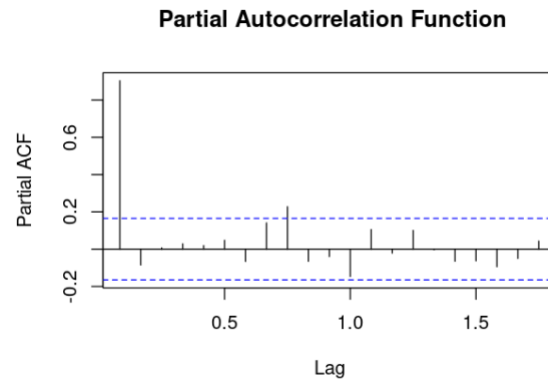


Figure 4 Partial Autocorrelation Function of Facebook Popularity

In *Figure 3*, the ACF plot is used to measure the correlation between the time series data and its lagged values over different time intervals where each bar represents the autocorrelation values for a specific lag. *Figure 3* shows that ACF decays slowly as the lag increases. This shows that there is an existence of seasonality or a trend in this time series data. *Figure 4* shows the correlation between the time series and its lagged values while having the influence of shorter lags being removed. In *Figure 4*, the first lag has the highest significance correlation, this proves that the time series has a strong direct relationship with its immediate past value. While on the other hand, the majority of the correlation is weaker and it falls within the confidence interval which indicates that they are not statistically significant. This proves that the time series data mainly depends on the most recent past observation at lag 1.

Hypothesis Test	Results
Augmented Dickey-Fuller Test	Augmented Dickey-Fuller Test <pre>data: train Dickey-Fuller = -2.9861, Lag order = 5, p-value = 0.166 alternative hypothesis: stationary</pre>
Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test	KPSS Test for Level Stationarity <pre>data: train KPSS Level = 1.1864, Truncation lag parameter = 4, p-value = 0.01</pre>

Table 1 Hypothesis Test of Facebook Popularity

The result of the ADF test showed that the test statistic value is -2.9861, the Lag order is 5 and the obtained p-value is 0.166, which is greater than the significance level, $\alpha = 0.05$. This means that the null hypothesis fails to be rejected. This proves that the time series has a unit root therefore the time series cannot be concluded as stationary, it could be non-stationary. The KPSS test is used to prove the time series is non-stationary. The result of the KPSS test showed that the KPSS level is 1.1864, Lag parameter is 4 and p-value is 0.01, which is less than the significance level, $\alpha = 0.05$. This means the null hypothesis is rejected and it is considered that the time

series is non-stationary. Based on both hypothesis tests, we can conclude that the time series is non-stationary.

3.2 Seasonal Differences

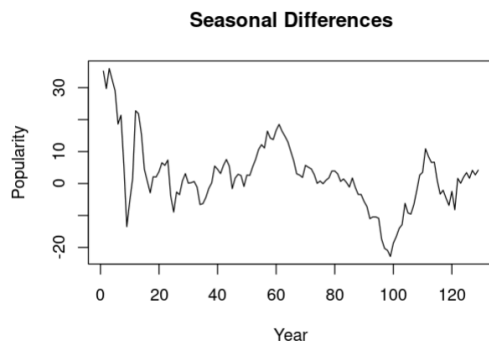


Figure 6 : Seasonal Differencing graph of Social Media Popularity

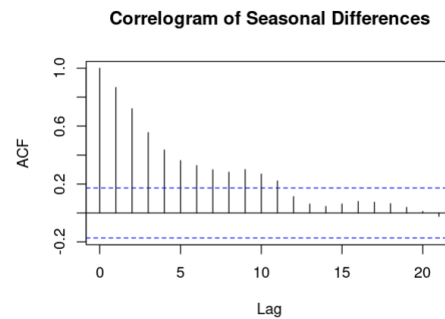


Figure 7 : ACF of Seasonal Differences

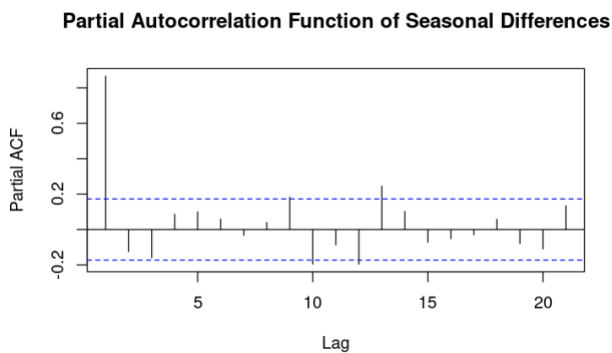


Figure 8 : PACF of Seasonal Differences

In *Figure 6*, after seasonal differencing, the time series plot appears to be stationary since there is no clear trend or pattern shown in the graph. This proves that the seasonal component has been removed. In *Figure 7*, the ACF plot shows that there might be some autocorrelation at higher lags which indicates that the first seasonal differences might not be entirely uncorrelated. In *Figure 8*, there could be a significant direct relationship between the current value and the previous value of the first seasonal differences due to the fact that the PACF plot shows a significant partial autocorrelation at lag 1. With that being said, there is a potential need for further stationarity testing. Besides that, we are not able to observe any spikes continuously on the correlogram which results that it is not a seasonality data.

Hypothesis Test	Results
Augmented Dickey-Fuller Test	Augmented Dickey-Fuller Test <pre>data: first_diff Dickey-Fuller = -2.9034, Lag order = 5, p-value = 0.201 alternative hypothesis: stationary</pre>

Phillips-Perron Unit Root Test	Phillips-Perron Unit Root Test data: first_diff Dickey-Fuller Z(alpha) = -19.699, Truncation lag parameter = 4, p-value = 0.06729 alternative hypothesis: stationary
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Table 2 : Hypothesis Test Conducted for Seasonal Differences

The result of the ADF test for the first seasonal differencing showed that the test statistic value is -2.9034, the Lag order is 5 and the obtained p-value is 0.201, which is greater than the significance level, $\alpha = 0.05$. Therefore the null hypothesis fails to be rejected, which indicates that we cannot conclude that the first seasonal differences are stationary based on the test. For the other test, the PP test result of the first seasonal differencing shows that the test statistic value is -19.699, the Truncation Lag Parameter is 4 and the obtained p-value is 0.06729, which is also greater than the 0.05. Therefore the null hypothesis failed to be rejected, which concluded that the data is non-stationarity. Based on both of the hypothesis test results, both indicate that the time series data that are tested using seasonal differencing are non-stationarity.

3.3 Non Seasonal Differences

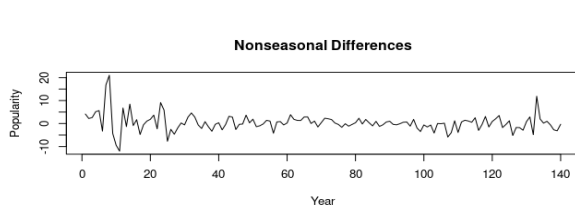


Figure 9 : Non-Seasonal Differences of Social Media Popularity

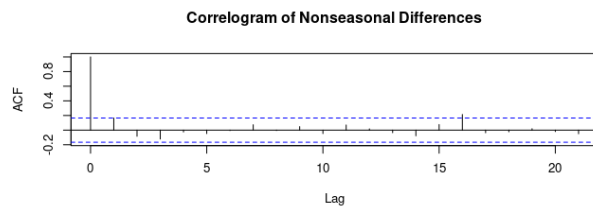


Figure 10 : ACF of NonSeasonal Differences

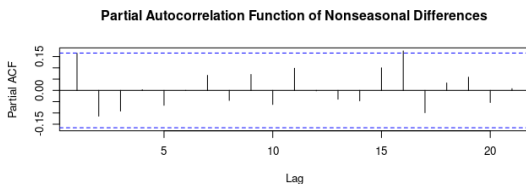


Figure 11 : PACF of NonSeasonal Differences

After non-seasonal differencing, *Figure 9* shows that the time series plot is stationary, this indicates that the non-seasonal component has been effectively removed. In *Figure 10*, there is no significant autocorrelation after the first lag. In *Figure 11*, there are no consistent, significant spikes at regular intervals which proves that there is no strong evidence of seasonality in the remaining data after going through non-seasonal differencing.

Hypothesis Test	Results
Augmented Dickey-Fuller Test	Augmented Dickey-Fuller Test data: second_diff Dickey-Fuller = -5.545, Lag order = 5, p-value = 0.01 alternative hypothesis: stationary

Phillips-Perron Unit Root Test	Phillips-Perron Unit Root Test data: second_diff Dickey-Fuller Z(alpha) = -103.68, Truncation lag parameter = 4, p-value = 0.01 alternative hypothesis: stationary
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Table 3 : Hypothesis Test for Non Seasonal Differences

The result for the ADF test of the second non-seasonal differencing showed that the test statistic value is -5.545, Lag order is 5 and p-value is 0.01, which is less than the significance level, $\alpha = 0.05$. Therefore the null hypothesis is rejected, which concludes that the data is stationarity. While the PP test result of the second non-seasonal differencing shows that the p-value is 0.01, which is also less than the 0.05. Therefore the null hypothesis is rejected, which indicates that the data is stationarity. Based on the 2 hypothesis test, we can conclude that the time series data differences using non seasonal differencing is stationarity.

3.4 Additive Decomposition

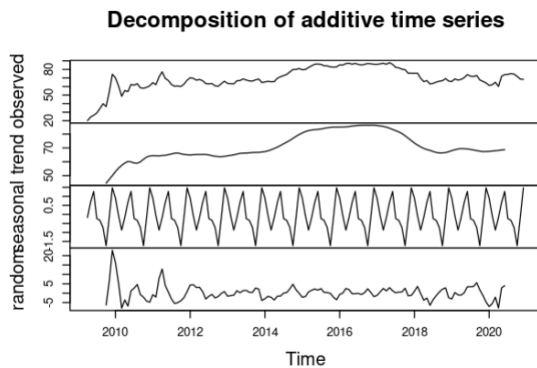


Figure 12 : Decomposition of Additive Time Series

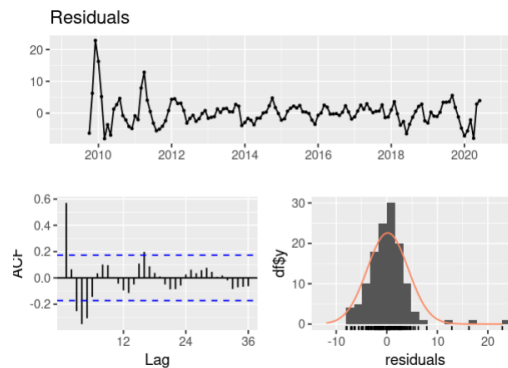


Figure 13 : Residuals plot of Additive Time Series

The decomposition in *Figure 12* effectively captured the trend and seasonal components of the time series. In *Figure 13*, the model is appropriate since the residuals seem to be random and normally distributed.

Hypothesis Test	Results
Ljung-Box test	Ljung-Box test data: Residuals Q* = 103.59, df = 24, p-value = 7.301e-12 Model df: 0. Total lags used: 24

Table 4 : Hypothesis Test for Additive Time Series

The result of the Ljung-Box test shows the Q-test is 103.59, df is 24 and p-value of 7.301e-12, which is less than the 0.05. Therefore, the null hypothesis is rejected, that there is evidence of significant autocorrelation in the residuals of the model. To make a conclusion for this, the model can effectively forecast future values of the time series.

4.0 Results and Discussion

4.1 Arima Model

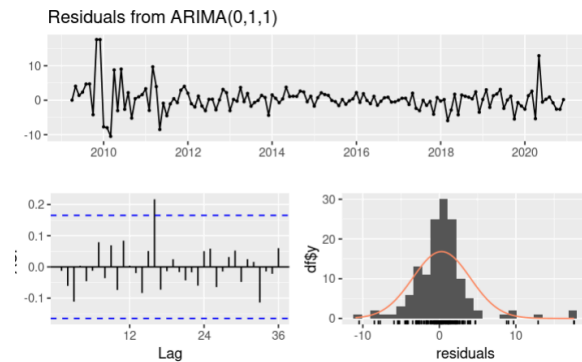


Figure 14 : Residual plot for Arima(0,1,1)

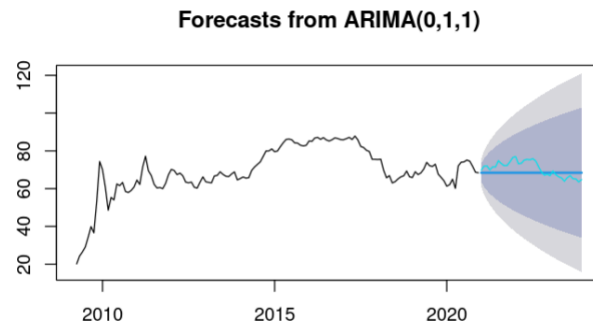


Figure 15 : Forecast graph for Arima(0,1,1)

Hypothesis Testing	Results
Ljung-Box Test	Ljung-Box test data: Residuals from ARIMA(0,1,1) Q* = 18.103, df = 23, p-value = 0.7519 Model df: 1. Total lags used: 24
Shapiro-Wilk test	Shapiro-Wilk normality test data: residuals_extract W = 0.85956, p-value = 2.945e-10

Table 5 : Hypothesis Testing of Arima(0,1,1)

The result of the Ljung-Box Test from the Residuals from ARIMA(0,1,1) is that the Q test is 18.103 and the p-value is 0.7519 which is greater than the 0.05. Thus, the null hypothesis fails to be rejected. This also indicates that the residual sequence of the model is white noise, there is no significant autocorrelation between the residuals. It indicates that the model has captured the main trends and patterns in the Facebook market. This means the ARIMA model performs well fitting the data. The results of the Shapiro-Wilk test, which is used to evaluate the Arima model, indicate that the residuals from the model depart from normality and have test statistics of 0.85956 and p-value of 2.945e-10, respectively. Based on *Figure 15*, the ARIMA model provides forecasts at the tail of the time series. The blue shaded area is the forecast interval in the *Figure 15 : Forecast graph for Arima(0,1,1)* becomes wider, indicating that the future uncertainty increases. This is because the social media market is greatly affected by external factors such as the changes in user behavior and the impact of it is hard to predict. The blue solid line is the future trend predicted by the model, showing a relatively stable forecast result. This is because the ARIMA model predicts future trends based on the last value and there is no obvious upward or downward trend in the last stage of historical data. So the model predicts that it will

remain in a stable state in the future which is Facebook's market share has stabilized in the near term.

4.2 ETS Model

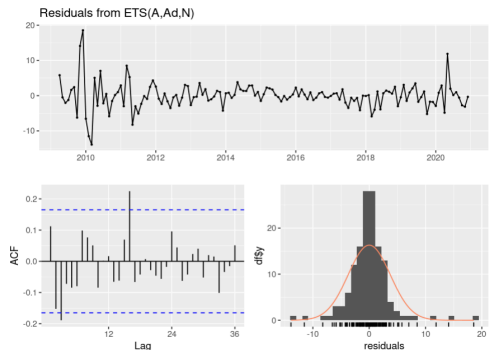


Figure 15 : Residual plot for ETS(A,AD,N)

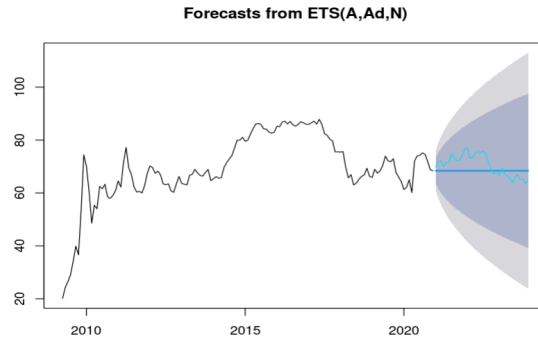


Figure 16 : Forecast graph for ETS(A,AD,N)

Hypothesis Testing	Results
Ljung-Box Test	Ljung-Box test data: Residuals from ETS(A,Ad,N) $Q^* = 30.9$, $df = 24$, $p\text{-value} = 0.1567$ Model df: 0. Total lags used: 24
Shapiro-Wilk test	Shapiro-Wilk normality test data: residuals_extract $W = 0.87742$, $p\text{-value} = 1.979e-09$

Table 6 : Hypothesis Testing for ETS Model

The Ljung-Box Test residuals of check is $Q^*=30.9$, $df=24$ and $p\text{-value}$ is 0.1567. Since the $p\text{-value}$ is greater than 0.05, thus the null hypothesis fails to be rejected. We can conclude that the residuals of the model are not significantly autocorrelated. It means that the model residuals conform to the characteristic of white noise, indicating that the model fits well and has effectively captured the pattern in data. Since the residuals is white noise, it means that the data trend of Facebook's market share has been effectively captured by the mode such as the upward or downward trend. The Shapiro-Wilk test that is used to test ETS model shows test statistics of 0.87742, with $p\text{-value}$ of $1.979e-09$ suggesting that the residuals from the Arima model that deviates from the Normal distribution and the residuals are less significantly difference to normality. Based on the Figure 16, ETS model is similar to the ARIMA model, the uncertainty of future forecasts increases over time but the actual line in the ARIMA model is more fit to the predicted line. Although the model fits the historical data well, the expansion of the blue shaded area shows the uncertainty in the future. The predicted line in Figure 16 is a straight horizontal line. This is due to the fact that the ETS model predicts based on the last value, the subsequent

predicted values will also be predicted based on the last values.

4.3 TBATS (Trigonometric, Box-Cox transformation, ARMA errors, Trend and Seasonal components)

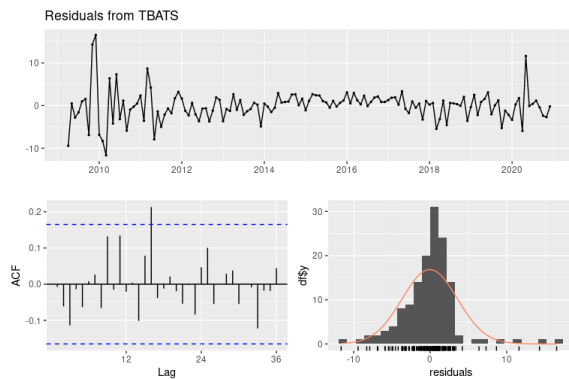


Figure 17 : Residuals plot for TBATS

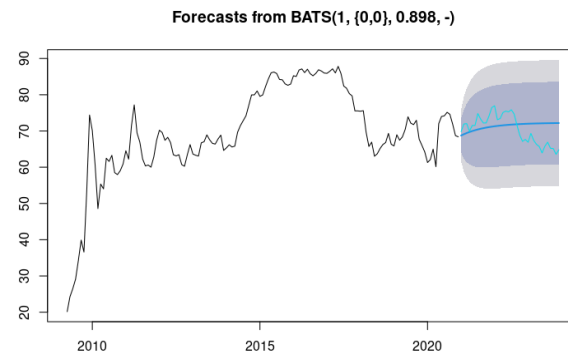


Figure 18 : Forecast graph for TBATS

Hypothesis Testing	Results
Ljung-Box Test	Ljung-Box test data: Residuals from TBATS $Q^* = 21.811$, $df = 24$, $p\text{-value} = 0.5906$ Model df: 0. Total lags used: 24
Shapiro-Wilk test	Shapiro-Wilk normality test data: residuals_extract $W = 0.88709$, $p\text{-value} = 5.979e-09$

Table 7 : Hypothesis Test of TBATS

The Ljung-Box Test residuals of check is $Q^*=21.811$, $df=24$ and $p\text{-value}$ is 0.5906. Since the $p\text{-value}$ is larger than 0.05, thus the null hypothesis fails to be rejected. We can conclude that the residuals of the model are not significantly autocorrelated. It means that the model residuals conform to the characteristic of white noise, indicating that the model fits well and has effectively captured the pattern in data. The TBATS model is evaluated using the Shapiro-Wilk test, and the test statistics of 0.88709 and 5.979e-09, respectively, show that the residuals from the model deviate from normality. Based on Figure 18, the upward slope of the blue line suggests a positive trend in the forecasted variable. It also has a relatively high degree of certainty based on the narrow prediction intervals.

5.0 Model Evaluation

Model	RMSE	Ljung-Box
ARIMA (0,1,1)	Training set : 3.751855 Testing set : 4.440683	$p\text{-value} = 0.7519$

ETS (A,Ad,N)	Training set : 3.719771 Test set : 4.456984	p-value = 0.1567
TBATS	Training set : 3.609572 Test set : 4.506730	p-value = 0.5906

Table 8 Compare all the model result evaluation

Based on *table 8*, We compared the RMSE values of the training set and test sets and the Ljung-Box test to find the best model. The RMSE difference between the training set and test is of the ARIMA model which is 0.68828, ETS model is 0.737213 and the TBATS model is 0.897158. By comparing these 3 differences, ARIMA has the smallest difference indicating that its performance is more stable. By comparing the result of the Ljung-Box test, the p-value of ARIMA is 0.7519, ETS is 0.1567 and 0.5906. The p-value of the ARIMA model is much higher than 0.05. This indicates that the residual sequence of the model can be considered as white noise, indicating that the model fits the data well and has no obvious autocorrelation. In summary, the ARIMA model performs well in both RMSE and Ljung-Box tests and is recommended as the best model.

```

arima(x = train, order = c(0, 1, 1))
Coefficients:
      ma1
      0.1940
s.e.  0.0838

sigma^2 estimated as 14.18:  log likelihood = -384.28,  aic = 772.57

Training set error measures:
      ME      RMSE      MAE      MPE      MAPE      MASE      ACF1
Training set 0.2864691 3.751855 2.376287 0.5501767 3.821807 0.9851062 -0.01237035

```

Figure : Summary of Best Fit Model, Arima (0,1,1)

The general form of Arima(p,q,r) is

$$(1-B)Y_t = (1 - \theta_1 B)\varepsilon_t$$

Y_t represents the observation value at time t , B is a backward operator that is $BY_t = Y_{t-1}$, which means the time series backward by one time unit. $(1-B)$ represents the difference operation on the time series with the purpose of making it stationary.

The equation for best fit model of Arima(0,1,1) is

$$Y_t = Y_{t-1} + \varepsilon_t - 0.1940\varepsilon_{t-1}$$

The existence of Y_t means that the number of Facebook users or market share at the current moment, Y_{t-1} indicates that there is a dependency between the current observation and the previous observation, which means that there is a trend in the time series. The ε_t part represents the current random fluctuation that affect the number of users or market share such market activities and changes in user preferences, while the $-0.1940\varepsilon_{t-1}$ indicates that the error of the

previous moment has a certain negative impact on the current value, indicating that the model takes the randomness of the previous moment into account when adjusting the forecast. In a social media environment, user behavior and platform performance are dynamic. For example, if Facebook's user growth is slow during a certain period, this phenomenon will be reflected in the current number of users through ϵ_{t-1} . The negative coefficient of -0.1940 means that if the early performance is poor, it will suppress the current number of users.

6.0 Conclusion

In modeling the share of social media, Facebook from April 2009 to December 2023, the best fit model is identified as the Arima(0,1,1) model with aic of 772.57, RMSE of 3.751855 for training set and 4.440683 for testing set which results in the lowest RMSE, AIC and p-value when using Ljung-Box Test compared to ETS(A, Ad,M) and TBATS model that has been proposed. In this analysis, the main trends and patterns of Facebook's monthly data is not captured even with the best model, the ARIMA model. This is because the forecast graph shows that the predicted line is a straight horizontal line. The ARIMA model will refer to the last value to predict future values, the prediction of the subsequent values will also refer to their respected previous values.

The forecast of the ARIMA model suggests that the variable. the share of Facebook is expected to remain constant. This would mean that the platform is experiencing limited growth or decline. This would imply that the business planning and resource allocation should be considered. The results of the study have important implications for business strategy and policy making. Companies can formulate corresponding marketing and product development strategies based on the trend forecasted by the ARIMA model. However, since the ARIMA model cannot capture the impact of external factors such as news on user behavior, this means that companies need to be cautious when making strategic decisions and cannot rely solely on model predictions. For the further research can explore the possibility of combining machine learning or other complex models to capture these sudden changes

The limitations of this analysis is that our best model, the ARIMA model, predicts the value based on the last value and it causes the predicted line to be a straight horizontal line. This would cause the underlying trend or pattern in the data not being captured effectively. This can be problematic when the time series exhibits non-linear relationships or significant external factors.

Given the complexity and dynamic characteristics of social media data, the GARCH model may be a more appropriate choice. It is able to capture the non-linear relationship in the data which is particularly important for social media data. This is because the data are often affected by various external factors such as the changes in user behavior, the rise of new social media platforms and others. In addition, these factors can also be incorporated into the model as additional predictors which could help capture the impact of these factors on the time series.

7.0 Reference

- Social media popularity (2009 - 2023). (2024, May 3). Kaggle.
<https://www.kaggle.com/datasets/michau96/social-media-popularity-2009-2023>
- Rehal, V., & Rehal, V. (2024, September 7). ARIMA and SARIMA in RStudio - SPUR ECONOMICS. SPUR ECONOMICS - Learn and Excel.
<https://spureconomics.com/arima-and-sarima-in-rstudio/>
- (Demographics of Social Media Users and Adoption in the United States, 2024)
<https://www.pewresearch.org/internet/fact-sheet/social-media/>

8.0 Appendices

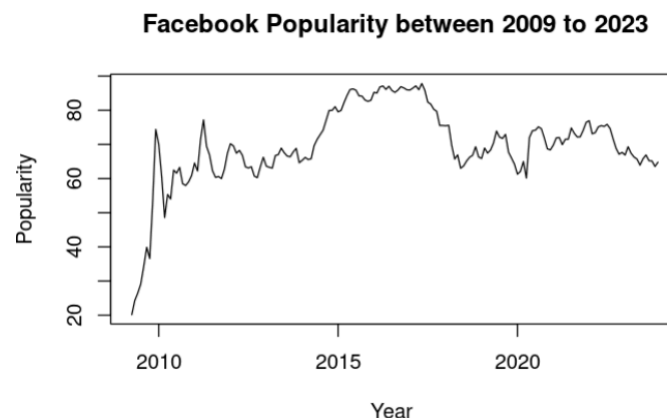
Import Datasets and Plot Overview Graph of the Dataset

```
# Import dataset
Social_media <- read.csv("social_media.csv")
SocialM <- Social_media[, c(1, 2)]
x <- SocialM[, 1]
y <- SocialM[, 2]

Data <- ts(y, frequency = 12, start = c(2009, 4), end = c(2023, 12))

## plot graph
plot(Data, xlab = "Year ", ylab = "Popularity", main = "Facebook Popularity between 2009 to 2023")
```

Graph :



Split Dataset into Training Set and Testing Set

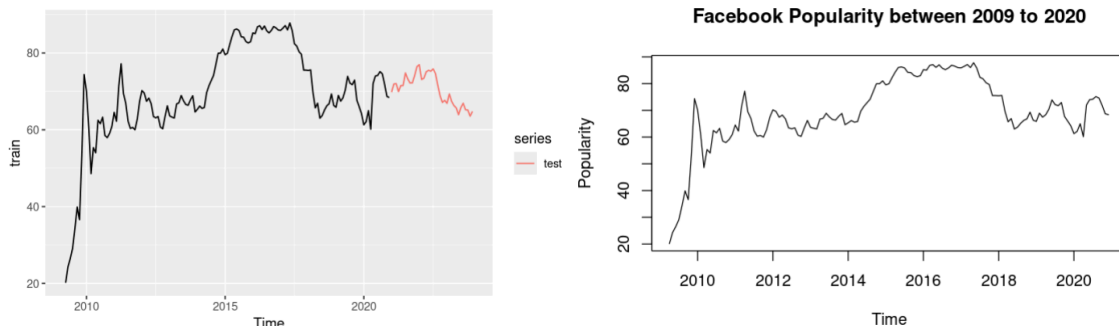
```
##-----
# Use train data from 2000 to 2019 for forecasting
train = window(Data, start=2009, end=c(2020,12));
train
# Use remaining data from 2020 to 2024 to test accuracy
test = window(Data, start=2021, end=c(2023,12)); test
autoplot(train)+autolayer(test)
traindt <- ts(train, frequency = 12, start = c(2009, 4), end = c(2020,12))

## plot graph
plot(traindt, ylab = "Popularity", main = "Facebook Popularity between 2009 to 2020")
```

Console :

```
> train
      Jan  Feb  Mar  Apr  May  Jun  Jul  Aug  Sep  Oct  Nov  Dec
2009      20.16 24.30 26.48 29.10 34.25 39.89 36.60 53.39 74.39
2010 70.08 60.61 48.59 55.35 54.03 62.48 61.60 63.28 58.51 57.97 59.05 60.88
2011 64.54 62.23 71.36 77.19 69.47 66.95 62.29 60.36 60.61 60.01 62.79 67.39
2012 70.21 69.58 67.43 68.25 66.81 63.45 63.12 63.46 60.72 60.27 63.43 66.24
2013 63.64 63.30 63.08 66.73 67.03 68.92 67.56 66.58 66.39 67.79 68.86 64.64
2014 65.34 66.19 65.58 65.81 69.66 71.47 72.84 74.18 77.05 79.95 79.97 81.04
2015 79.55 79.95 82.25 84.32 86.05 86.26 85.86 84.22 84.13 83.00 82.61 82.95
2016 85.24 85.04 86.82 87.12 86.10 87.04 85.80 85.23 85.89 86.90 86.54 85.98
2017 85.90 86.47 87.15 86.02 87.83 85.87 82.39 81.80 80.31 79.68 75.54 75.54
2018 75.45 75.60 69.72 65.73 66.92 63.03 63.71 65.15 66.25 66.81 69.33 66.33
2019 65.88 68.95 67.46 68.36 70.42 73.91 72.16 71.75 72.94 67.73 66.01 64.22
2020 61.30 62.11 64.99 60.15 72.02 74.00 74.17 75.15 74.59 71.85 68.72 68.39
> # Use remaining data from 2020 to 2024 to test accuracy
> test = window(Data, start=2021, end=c(2023,12)); test
      Jan  Feb  Mar  Apr  May  Jun  Jul  Aug  Sep  Oct  Nov  Dec
2021 69.80 71.92 72.03 69.94 71.51 71.53 74.81 73.27 72.19 72.22 74.13 76.47
2022 76.95 73.12 73.44 75.10 75.49 75.27 75.86 74.65 71.64 68.87 67.13 67.67
2023 66.90 69.33 67.41 66.31 65.70 63.94 65.78 66.91 65.24 65.15 63.56 64.79
```

Graph :



Auto Arima Model

```
##-----  
## Auto Arima  
library(forecast)  
arima_fit <- auto.arima(train, ic="aic", trace=TRUE)  
summary(arima_fit)  
checkresiduals(arima_fit)  
fr <- forecast(arima_fit, h=length(test))  
layout(matrix(c(1,1)))  
plot(fr)  
lines(test, col="turquoise2")  
accuracy(fr, test)
```

Console :

```
> ## Auto Arima  
> library(forecast)  
> arima_fit <- auto.arima(train, ic="aic", trace=TRUE)  
  
ARIMA(2,1,2)(1,0,1)[12] with drift : 780.8477  
ARIMA(0,1,0) with drift : 776.2609  
ARIMA(1,1,0)(1,0,0)[12] with drift : 776.3603  
ARIMA(0,1,1)(0,0,1)[12] with drift : 775.6958  
ARIMA(0,1,0) : 775.3974  
ARIMA(0,1,0)(1,0,0)[12] with drift : 778.1709  
ARIMA(0,1,0)(0,0,1)[12] with drift : 778.1872  
ARIMA(0,1,0)(1,0,1)[12] with drift : 779.7142  
ARIMA(1,1,0) with drift : 774.4229  
ARIMA(1,1,0)(0,0,1)[12] with drift : 776.3723  
ARIMA(1,1,0)(1,0,1)[12] with drift : 777.8022  
ARIMA(2,1,0) with drift : 774.6039  
ARIMA(1,1,1) with drift : 775.6124  
ARIMA(0,1,1) with drift : 773.7155  
ARIMA(0,1,1)(1,0,0)[12] with drift : 775.6912  
ARIMA(0,1,1)(1,0,1)[12] with drift : 777.2337  
ARIMA(0,1,2) with drift : 775.4587  
ARIMA(1,1,2) with drift : 776.2  
ARIMA(0,1,1) : 772.5677  
ARIMA(0,1,1)(1,0,0)[12] : 774.5367  
ARIMA(0,1,1)(0,0,1)[12] : 774.5424  
ARIMA(0,1,1)(1,0,1)[12] : 776.0128  
ARIMA(1,1,1) : 774.4942  
ARIMA(0,1,2) : 774.3957  
ARIMA(1,1,0) : 773.2568  
ARIMA(1,1,2) : 775.4495  
  
Best model: ARIMA(0,1,1)  
  
> summary(arima_fit)  
Series: train  
ARIMA(0,1,1)  
  
Coefficients:  
ma1  
0.1940  
s.e. 0.0838
```

```

sigma^2 = 14.28: log likelihood = -384.28
AIC=772.57 AICc=772.66 BIC=778.45

Training set error measures:
      ME      RMSE      MAE      MPE      MAPE      MASE      ACF1
Training set 0.2864691 3.751855 2.376287 0.5501767 3.821807 0.3140275 -0.01237035
> checkresiduals(arima_fit)

Ljung-Box test

data: Residuals from ARIMA(0,1,1)
Q* = 18.103, df = 23, p-value = 0.7519

Model df: 1. Total lags used: 24

> fr <- forecast(arima_fit, h=length(test))
> layout(matrix(c(1,1)))
> plot(fr)
> lines(test, col="turquoise2")
> accuracy(fr, test)
      ME      RMSE      MAE      MPE      MAPE      MASE      ACF1 Theil's U
Training set 0.2864691 3.751855 2.376287 0.5501767 3.821807 0.3140275 -0.01237035 NA
Test set      2.0204771 4.440683 3.855611 2.5585467 5.371678 0.5095208 0.88156373 2.593112

> checkresiduals(arima_fit)

Ljung-Box test

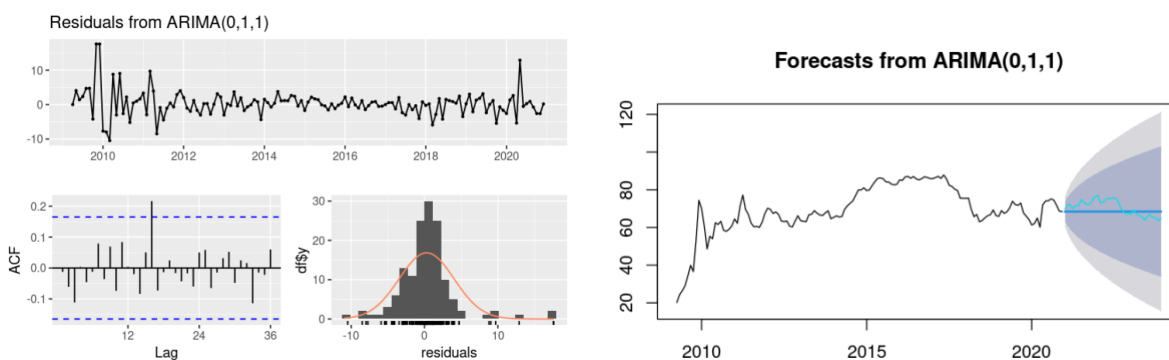
data: Residuals from ARIMA(0,1,1)
Q* = 18.103, df = 23, p-value = 0.7519

Model df: 1. Total lags used: 24

> fr <- forecast(arima_fit, h=length(test))
> layout(matrix(c(1,1)))
> plot(fr)
> lines(test, col="turquoise2")
> accuracy(fr, test)
      ME      RMSE      MAE      MPE      MAPE      MASE      ACF1
Training set 0.2864691 3.751855 2.376287 0.5501767 3.821807 0.3140275 -0.01237035
Test set      2.0204771 4.440683 3.855611 2.5585467 5.371678 0.5095208 0.88156373
Theil's U
Training set NA
Test set      2.593112

```

Graph :



Autocorrelation Function and Partial Autocorrelation Function of Facebook Popularity

```
##-----
library(tseries)

# Autocorrelation Function
acf(train, main = "Autocorrelation Function")

# Partial Autocorrelation Function
pacf(train, main = " Partial Autocorrelation Function")

adf.test(train) # model is non-stationary
kpss.test(train) # model is non-stationary
```

Console :

```
> adf.test(train) # model is non-stationary

Augmented Dickey-Fuller Test

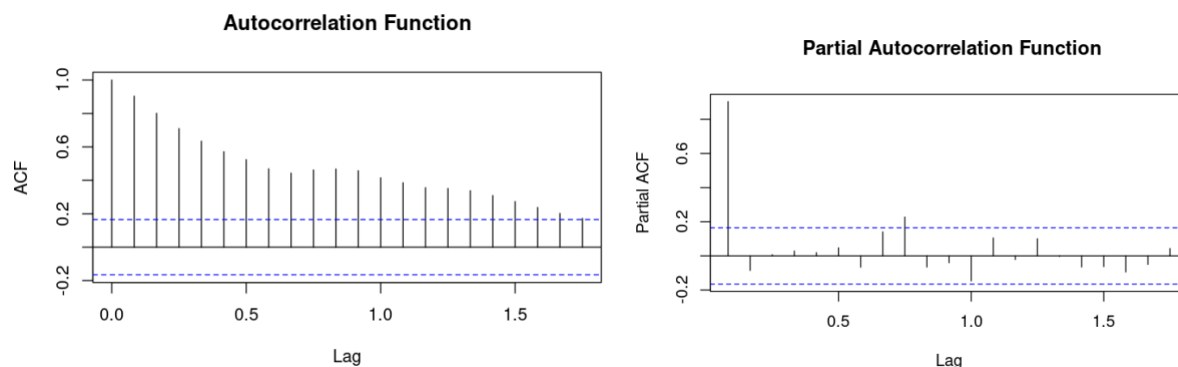
data: train
Dickey-Fuller = -2.9861, Lag order = 5, p-value = 0.166
alternative hypothesis: stationary

> kpss.test(train) # model is non-stationary

KPSS Test for Level Stationarity

data: train
KPSS Level = 1.1864, Truncation lag parameter = 4, p-value = 0.01
```

Graph:



Additive Decomposition

```
##-----
## Decompostion using Additive - it is a seasonal model
decomposed_comp <- decompose(train, type="additive")
trend <- decomposed_comp$trend
seasonal <- decomposed_comp$seasonal
remainder <- decomposed_comp$random
plot(decomposed_comp)
checkresiduals(remainder)
cbind(decomposed_comp$x,decomposed_comp$trend,decomposed_comp$seasonal,decomposed_comp$random)
```

Console :

```
> checkresiduals(remainder)
```

Ljung-Box test

data: Residuals

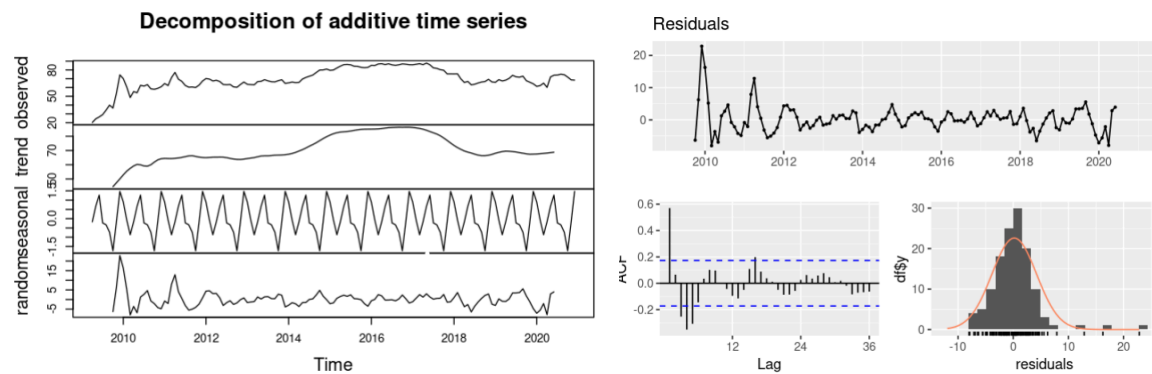
Q* = 103.59, df = 24, p-value = 7.301e-12

Model df: 0. Total lags used: 24

	decomposed_comp\$x	decomposed_comp\$trend	decomposed_comp\$seasonal	decomposed_comp\$random
Apr 2009	20.16	NA	-0.16994318	NA
May 2009	24.30	NA	0.66664773	NA
Jun 2009	26.48	NA	1.27130682	NA
Jul 2009	29.10	NA	-0.23074621	NA
Aug 2009	34.25	NA	-0.33241288	NA
Sep 2009	39.89	NA	-0.72799621	NA
Oct 2009	36.60	44.61958	-1.73289773	-6.28668561
Nov 2009	53.39	47.32458	-0.17331439	6.23873106
Dec 2009	74.39	50.06333	1.47320076	22.85346591
Jan 2010	70.08	52.91750	0.89793561	16.26456439
Feb 2010	60.61	55.48125	-0.07316288	5.20191288
Mar 2010	48.59	57.46667	-0.86861742	-8.00804924
Apr 2010	55.35	59.13292	-0.16994318	-3.61297348
May 2010	54.03	60.25917	0.66664773	-6.89581439
Jun 2010	62.48	59.93208	1.27130682	1.27660985
Jul 2010	61.60	59.13833	-0.23074621	2.69241288
Aug 2010	63.28	58.97500	-0.33241288	4.63741288
Sep 2010	58.51	59.99125	-0.72799621	-0.75325379
Oct 2010	57.97	61.85000	-1.73289773	-2.14710227
Nov 2010	59.05	63.40333	-0.17331439	-4.18001894
Dec 2010	60.88	64.23292	1.47320076	-4.82611742
Jan 2011	64.54	64.44792	0.89793561	-0.80585227
Feb 2011	62.23	64.35500	-0.07316288	-2.05183712
Mar 2011	71.36	64.32083	-0.86861742	7.90778409
Apr 2011	77.19	64.49333	-0.16994318	12.86660985
May 2011	69.47	64.73417	0.66664773	4.06918561
Jun 2011	66.95	65.16125	1.27130682	0.51744318
Jul 2011	62.29	65.66875	-0.23074621	-3.14800379
Aug 2011	60.36	66.21125	-0.33241288	-5.51883712
Sep 2011	60.61	66.35375	-0.72799621	-5.01575379
Oct 2011	60.01	65.81750	-1.73289773	-4.07460227
Nov 2011	62.79	65.33417	-0.17331439	-2.37085227
Dec 2011	67.39	65.07750	1.47320076	0.83929924
Jan 2012	70.21	64.96625	0.89793561	4.34581439
Feb 2012	69.58	65.13000	-0.07316288	4.52316288
Mar 2012	67.43	65.26375	-0.86861742	3.03486742
Apr 2012	68.25	65.27917	-0.16994318	3.14077652
May 2012	66.81	65.31667	0.66664773	0.82668561
Jun 2012	63.45	65.29542	1.27130682	-3.11672348
Jul 2012	63.12	64.97375	-0.23074621	-1.62300379
Aug 2012	63.46	64.43833	-0.33241288	-0.64592045
Sep 2012	60.72	63.99542	-0.72799621	-2.54742045
Oct 2012	60.27	63.75083	-1.73289773	-1.74793561
Nov 2012	63.43	63.69667	-0.17331439	-0.09335227
Dec 2012	66.24	63.93375	1.47320076	0.83304924
Jan 2013	63.64	64.34667	0.89793561	-1.60460227
Feb 2013	63.30	64.66167	-0.07316288	-1.28850379
Mar 2013	63.08	65.02792	-0.86861742	-1.07929924
Apr 2013	66.73	65.57750	-0.16994318	1.32244318
May 2013	67.03	66.11708	0.66664773	0.24626894
Jun 2013	68.92	66.27667	1.27130682	1.37202652
Jul 2013	67.56	66.28083	-0.23074621	1.50991288
Aug 2013	66.58	66.47208	-0.33241288	0.44032955
Sep 2013	66.39	66.69667	-0.72799621	0.42132955
Oct 2013	67.79	66.76250	-1.73289773	2.76039773
Nov 2013	68.86	66.83375	-0.17331439	2.19956439
Dec 2013	64.64	67.04958	1.47320076	-3.88278409
Jan 2014	65.34	67.37583	0.89793561	-2.93376894
Feb 2014	66.19	67.91250	-0.07316288	-1.64933712
Mar 2014	65.58	68.67333	-0.86861742	-2.22471591
Apr 2014	65.81	69.62417	-0.16994318	-3.64422348
May 2014	69.66	70.59375	0.66664773	-1.60039773
Jun 2014	71.47	71.74000	1.27130682	-1.54130682
Jul 2014	72.84	73.01542	-0.23074621	0.05532955
Aug 2014	74.18	74.18083	-0.33241288	0.33157955
Sep 2014	77.05	75.44875	-0.72799621	2.32924621
Oct 2014	79.95	76.91458	-1.73289773	4.76831439
Nov 2014	79.97	78.36875	-0.17331439	1.77456439
Dec 2014	81.04	79.66792	1.47320076	-0.10111742
Jan 2015	79.55	80.82667	0.89793561	-2.17460227
Feb 2015	79.95	81.78750	-0.07316288	-1.76433712
Mar 2015	82.25	82.50083	-0.86861742	0.61778409
Apr 2015	84.32	82.92292	-0.16994318	1.56702652
May 2015	86.05	83.16000	0.66664773	2.22335227
Jun 2015	86.26	83.34958	1.27130682	1.63910985
Jul 2015	85.86	83.66625	-0.23074621	2.42449621
Aug 2015	84.22	84.11542	-0.33241288	0.43699621
Sep 2015	84.13	84.51792	-0.72799621	0.34007955
Oct 2015	83.00	84.82500	-1.73289773	-0.09210227
Nov 2015	82.61	84.94375	-0.17331439	-2.16043561
Dec 2015	82.95	84.97833	1.47320076	-3.50153409
Jan 2016	85.24	85.00833	0.89793561	-0.66626894
Feb 2016	85.04	85.04792	-0.07316288	0.06524621
Mar 2016	86.82	85.16333	-0.86861742	2.52528409

Apr 2016	87.12	85.39917	-0.16994318	1.89077652
May 2016	86.10	85.72542	0.66664773	-0.29206439
Jun 2016	87.04	86.01542	1.27130682	-0.24672348
Jul 2016	85.80	86.16917	-0.23074621	-0.13842045
Aug 2016	85.23	86.25625	-0.33241288	-0.69383712
Sep 2016	85.89	86.32958	-0.72799621	0.28841288
Oct 2016	86.90	86.29750	-1.73289773	2.33539773
Nov 2016	86.54	86.32375	-0.17331439	0.38956439
Dec 2016	85.98	86.34708	1.47320076	-1.84028409
Jan 2017	85.90	86.15625	0.89793561	-1.15418561
Feb 2017	86.47	85.87125	-0.07316288	0.67191288
Mar 2017	87.15	85.49583	-0.86861742	2.52278409
Apr 2017	86.02	84.96250	-0.16994318	1.22744318
May 2017	87.83	84.20333	0.66664773	2.96001894
Jun 2017	85.87	83.31000	1.27130682	1.28869318
Jul 2017	82.39	82.43958	-0.23074621	0.18116288
Aug 2017	81.80	81.55125	-0.33241288	0.58116288
Sep 2017	80.31	80.37208	-0.72799621	0.66591288
Oct 2017	79.68	78.80042	-1.73289773	2.61248106
Nov 2017	75.54	77.08375	-0.17331439	-1.37043561
Dec 2017	75.54	75.26083	1.47320076	-1.19403409
Jan 2018	75.45	73.53083	0.89793561	1.02123106
Feb 2018	75.60	72.05875	-0.07316288	3.61441288
Mar 2018	69.72	70.77917	-0.86861742	-0.19054924
Apr 2018	65.73	69.65708	-0.16994318	-3.75714015
May 2018	66.92	68.86208	0.66664773	-2.60873106
Jun 2018	63.03	68.21958	1.27130682	-6.46089015
Jul 2018	63.71	67.43708	-0.23074621	-3.49633712
Aug 2018	65.15	66.76125	-0.33241288	-1.27883712
Sep 2018	66.25	66.39000	-0.72799621	0.58799621
Oct 2018	66.81	66.40542	-1.73289773	2.13748106
Nov 2018	69.33	66.66083	-0.17331439	2.84248106
Dec 2018	66.33	67.26000	1.47320076	-2.40320076
Jan 2019	65.88	68.06542	0.89793561	-3.08335227
Feb 2019	68.95	68.69250	-0.07316288	0.33066288
Mar 2019	67.46	69.24625	-0.86861742	-0.91763258
Apr 2019	68.36	69.56333	-0.16994318	-1.03339015
May 2019	70.42	69.46333	0.66664773	0.29001894
Jun 2019	73.91	69.23708	1.27130682	3.40160985
Jul 2019	72.16	68.95833	-0.23074621	3.43241288
Aug 2019	71.75	68.48250	-0.33241288	3.59991288
Sep 2019	72.94	68.09458	-0.72799621	5.57341288
Oct 2019	67.73	67.64958	-1.73289773	1.81331439
Nov 2019	66.01	67.37417	-0.17331439	-1.19085227
Dec 2019	64.22	67.44458	1.47320076	-4.69778409
Jan 2020	61.30	67.53208	0.89793561	-7.13001894
Feb 2020	62.11	67.75750	-0.07316288	-5.57433712
Mar 2020	64.99	67.96792	-0.86861742	-2.10929924
Apr 2020	60.15	68.20833	-0.16994318	-7.88839015
May 2020	72.02	68.49292	0.66664773	2.86043561
Jun 2020	74.00	68.77958	1.27130682	3.94910985
Jul 2020	74.17	NA	-0.23074621	NA
Aug 2020	75.15	NA	-0.33241288	NA
Sep 2020	74.59	NA	-0.72799621	NA
Oct 2020	71.85	NA	-1.73289773	NA
Nov 2020	68.72	NA	-0.17331439	NA
Dec 2020	68.39	NA	1.47320076	NA

Graph :



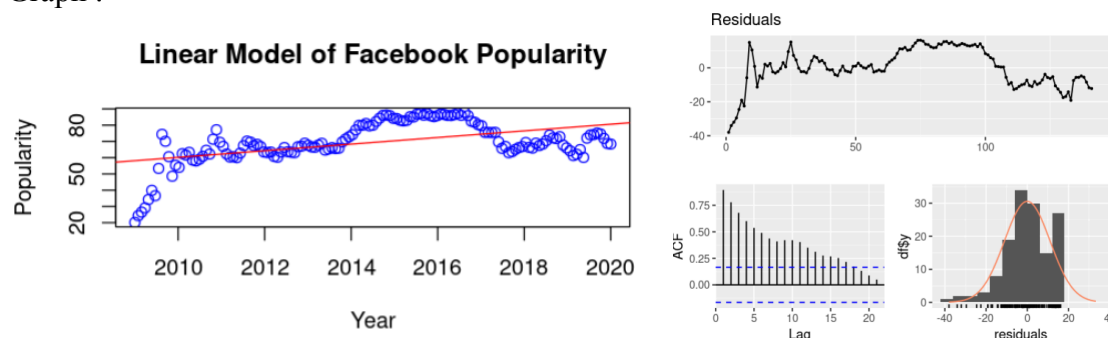
Linear Model

```
##-----  
# Define response variable and time variable for the linear model  
Y_linear <- train  
X_train <- seq(from = 2009, to = 2020, length.out = length(train))  
print(X_train)  
  
# Plot the training data with linear model  
plot(X_train, Y_linear, col = "blue", xlab = "Year", ylab = "Popularity", main = "Linear Model of Facebook Popularity")  
trend <- lm(Y_linear ~ X_train)  
  
# Add the regression line to the plot  
abline(trend, col = "red")  
  
# Summary of the linear model  
summary(trend) # significant positive trend  
  
# Check residuals of the linear model  
checkresiduals(trend)
```

Console :

```
Call:  
lm(formula = Y_linear ~ X_train)  
  
Residuals:  
    Min       1Q   Median       3Q      Max   
-38.006  -5.898   0.445  10.325  16.233  
  
Coefficients:  
            Estimate Std. Error t value Pr(>|t|)      
(Intercept) -4039.8718    590.7364  -6.839 2.33e-10 ***  
X_train         2.0398      0.2932   6.956 1.26e-10 ***  
---  
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
  
Residual standard error: 11.14 on 139 degrees of freedom  
Multiple R-squared:  0.2582,    Adjusted R-squared:  0.2529   
F-statistic: 48.39 on 1 and 139 DF,  p-value: 1.256e-10  
  
>  
> # Check residuals of the linear model  
> checkresiduals(trend)  
  
Breusch-Godfrey test for serial correlation of order up to 10  
  
data: Residuals  
LM test = 116.52, df = 10, p-value < 2.2e-16
```

Graph :



Explanation :

The scatter plot in *Figure 5* displays a positive linear relationship between the year and Facebook's popularity, whereas the year increases, so does the popularity. In *Figure 6*, the residuals plot shows no clear pattern which suggests that the assumption of equal or similar variances in different groups being compared is reasonably met. In the ACF plot, it shows there might be some correlation in residuals at higher lags. The reason for that is the lack of fit

or any sort of omitted variables. The residuals are approximately normally distributed after looking at the histogram and density plot although they might be some outliers. In Table 2, the overall model is statistically significant as it is indicated by the F-statistic, 48.39 and its p-value, 1.256e-10. This would mean that the predictor variable, year significantly explains the variation in the target variable, popularity. The coefficient for the variable, year which is 0.16027 with the p-value of 1.26e-10 is statistically significant which means year is a significant predictor of the target variable, popularity of Facebook. The value of R-squared shows that about 25.82% of the variation in the popularity variable is explained by the model, which means the model is not a perfect fit but still provides some explanatory power. In Table 3, the Breusch-Godfrey test is used to detect serial correlation in the residuals of a regression model. This is to confirm whether the assumption of independence in linear regression is or is not violated due to the occurrence of serial correlation when the residuals are correlated with each other. Since the p-value of the test is significantly less than $\alpha = 0.05$, the null hypothesis will be rejected and it can be concluded that there is strong evidence of serial correlation in the residuals of the linear model. Serial correlation could lead to some issues such as incorrect inferences about the model's parameters, inefficient estimates and also inaccurate standard errors.

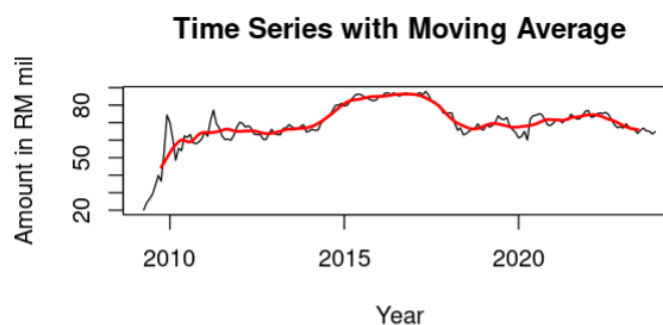
Moving Average

```
##-----
# Calculate the moving average with a window size of 12
movavg <- ma(Data, order = 12)

# Plot the original time series
plot(Data,
      main = "Time Series with Moving Average", |
      ylab = "Amount in RM mil",
      xlab = "Year")

# Add the moving average line to the plot
lines(movavg, col = "red", lwd = 2)
```

Graph :



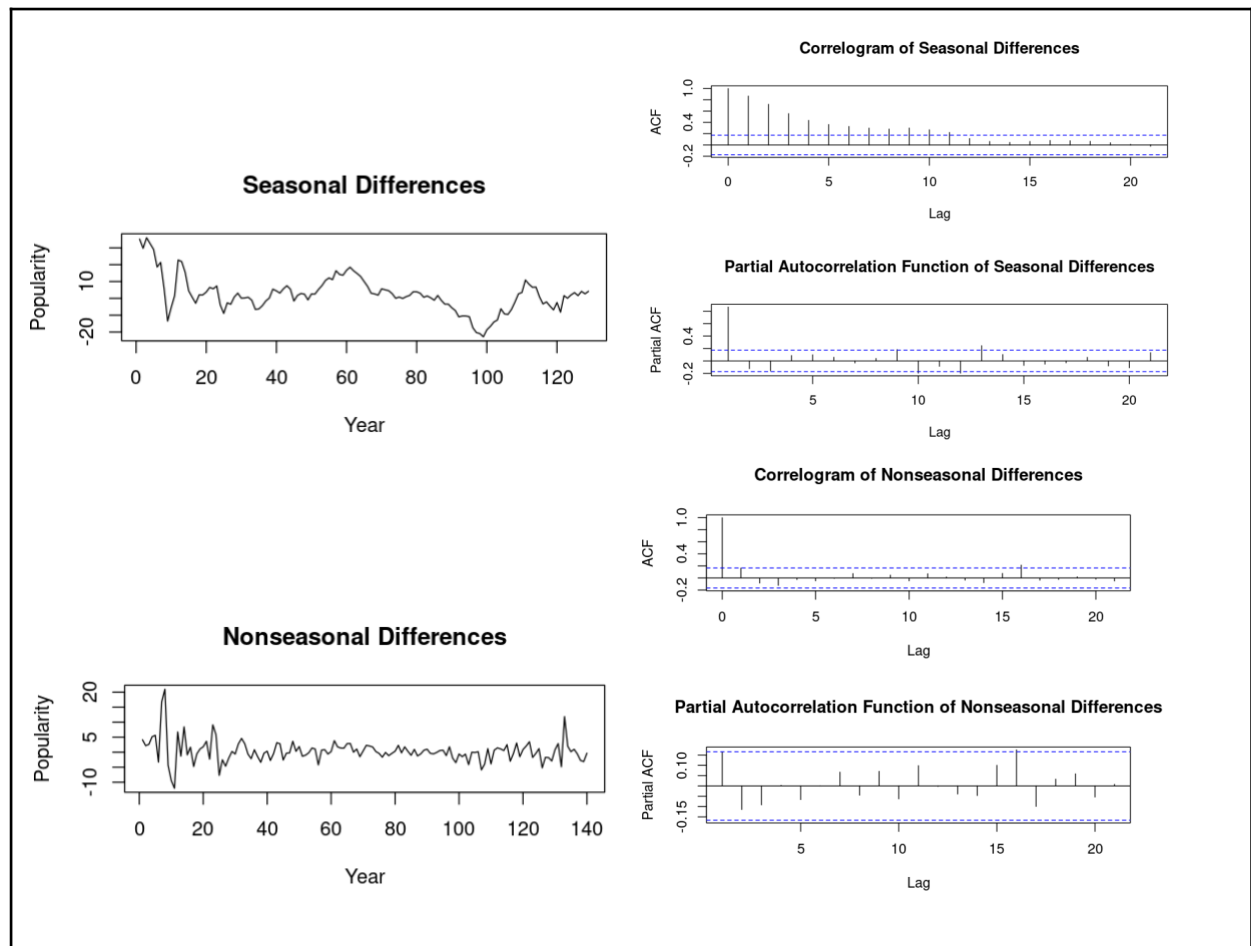
Seasonal Differencing & Non Seasonal Differencing

```
##-----  
first_diff <- ts(diff(train, lag=12))  
ts.plot(first_diff, gpars=list(main= "Seasonal Differences", xlab="Year",  
                               ylab="Popularity", lty=1))  
acf(first_diff, main="Correlogram of Seasonal Differences")  
pacf(first_diff, main="Partial Autocorrelation Function of Seasonal Differences")  
  
adf.test(first_diff)  
pp.test(first_diff)  
  
second_diff <- ts(diff(train, lag=1))  
ts.plot(second_diff, gpars=list(main= "Nonseasonal Differences", xlab="Year",  
                                ylab="Popularity", lty=1))  
  
acf(second_diff, main="Correlogram of Nonseasonal Differences")  
pacf(second_diff, main="Partial Autocorrelation Function of Nonseasonal Differences")  
  
adf.test(second_diff)  
pp.test(second_diff)
```

Console :

```
> adf.test(first_diff)  
  
      Augmented Dickey-Fuller Test  
  
data: first_diff  
Dickey-Fuller = -2.9034, Lag order = 5, p-value = 0.201  
alternative hypothesis: stationary  
  
> pp.test(first_diff)  
  
      Phillips-Perron Unit Root Test  
  
data: first_diff  
Dickey-Fuller Z(alpha) = -19.699, Truncation lag parameter = 4, p-value = 0.06729  
alternative hypothesis: stationary  
  
      Augmented Dickey-Fuller Test  
  
data: second_diff  
Dickey-Fuller = -5.545, Lag order = 5, p-value = 0.01  
alternative hypothesis: stationary  
  
Warning message:  
In adf.test(second_diff) : p-value smaller than printed p-value  
> pp.test(second_diff)  
  
      Phillips-Perron Unit Root Test  
  
data: second_diff  
Dickey-Fuller Z(alpha) = -103.68, Truncation lag parameter = 4, p-value = 0.01  
alternative hypothesis: stationary  
  
Warning message:  
In pp.test(second_diff) : p-value smaller than printed p-value
```

Graph :



Arima Model

```
##-----
## Arima

arimafit <- arima(train, order=c(0,1,1));
summary(arimafit)
checkresiduals(arimafit)
fr <- forecast(arimafit, h=length(test))
layout(matrix(c(1,1)))
plot(fr)
lines(test, col="turquoise2")
accuracy(fr, test)

residuals_extract <- residuals(arimafit)

## Shapiro Test
arima_sw <- shapiro.test(residuals_extract)
arima_sw

## Jarque Bera test
arima_jbt <- jarque.bera.test(residuals_extract)
arima_jbt

# Calculate AIC and BIC
aic_arima <- AIC(arimafit)
aic_arima

bic_arima <- BIC(arimafit)
bic_arima
```

Console :

```

Call:
arima(x = train, order = c(0, 1, 1))

Coefficients:
      mal 
0.1940 
s.e.  0.0838 

sigma^2 estimated as 14.18:  log likelihood = -384.28,  aic = 772.57

Training set error measures:
      ME      RMSE      MAE      MPE      MAPE      MASE      ACF1
Training set 0.2864691 3.751855 2.376287 0.5501767 3.821807 0.9851062 -0.01237035
> checkresiduals(arimafit)

      Ljung-Box test

data:  Residuals from ARIMA(0,1,1)
Q* = 18.103, df = 23, p-value = 0.7519

Model df: 1.   Total lags used: 24

> fr <- forecast(arimafit, h=length(test))
> layout(matrix(c(1,1)))
> plot(fr)
> lines(test, col="turquoise2")
> accuracy(fr, test)
      ME      RMSE      MAE      MPE      MAPE      MASE      ACF1
Training set 0.2864691 3.751855 2.376287 0.5501767 3.821807 0.3140275 -0.01237035
Test set     2.0204771 4.440683 3.855611 2.5585467 5.371678 0.5095208  0.88156373
Theil's U
Training set      NA
Test set         2.593112

> ## Shapiro Test
> arima_sw <- shapiro.test(residuals_extract)
> arima_sw

      Shapiro-Wilk normality test

data:  residuals_extract
W = 0.85956, p-value = 2.945e-10

>
> ## Jarque bera test
> arima_jbt <- jarque.bera.test(residuals_extract)
> arima_jbt

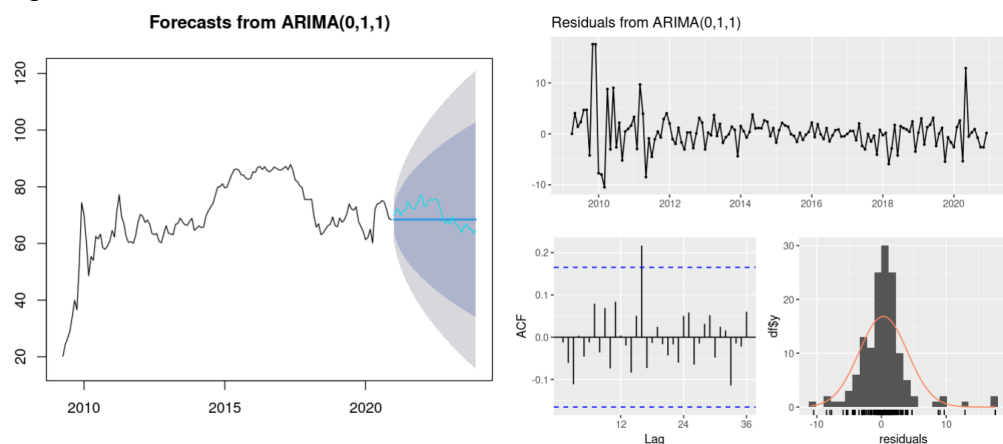
      Jarque Bera Test

data:  residuals_extract
X-squared = 294.78, df = 2, p-value < 2.2e-16

>
> # Calculate AIC and BIC
> aic_arima <- AIC(arimafit)
> aic_arima
[1] 772.5677
>
> bic_arima <- BIC(arimafit)
> bic_arima
[1] 778.451

```

Graph :



ETS Model

```
##-----  
#ETS model  
ets_fit <- ets(train)  
summary(ets_fit)  
checkresiduals(ets_fit)  
  
## Forecasting  
fr <- forecast(ets_fit, h=length(test))  
plot(fr)  
lines(test, col="turquoise2")  
accuracy(fr, test)  
  
residuals_extract <- residuals(ets_fit)  
  
## Shapiro Test  
ets_sw <- shapiro.test(residuals_extract)  
ets_sw  
  
## Jarque bera test  
ets_jbt <-jarque.bera.test(residuals_extract)  
ets_jbt  
  
# Calculate AIC and BIC  
aic_ets <- AIC(ets_fit)  
aic_ets  
bic_ets <-BIC(ets_fit)  
bic_ets
```

Console :

```
Call:
ets(y = train)

Smoothing parameters:
  alpha = 0.9999
  beta  = 1e-04
  phi   = 0.9131

Initial states:
  l = 9.2843
  b = 5.5498

sigma: 3.7875

      AIC      AICc      BIC
1080.228 1080.855 1097.920

Training set error measures:
              ME      RMSE      MAE      MPE      MAPE      MASE      ACF1
Training set 0.005503801 3.719771 2.375875 -0.03907017 3.856681 0.3139730 0.1120764
> checkresiduals(ets_fit)

Ljung-Box test

data: Residuals from ETS(A,Ad,N)
Q* = 30.9, df = 24, p-value = 0.1567

Model df: 0. Total lags used: 24

> accuracy(tr, test)
              ME      RMSE      MAE      MPE      MAPE      MASE      ACF1
Training set 0.005503801 3.719771 2.375875 -0.03907017 3.856681 0.3139730 0.1120764
Test set     2.056390994 4.456984 3.865346  2.60970512 5.383086 0.5108073 0.8815556
      Theil's U
Training set      NA
Test set         2.601117
> residuals_extract <- residuals(ets_fit)
>
> ## Shapiro Test
> ets_sw <- shapiro.test(residuals_extract)
> ets_sw

Shapiro-Wilk normality test

data: residuals_extract
W = 0.87742, p-value = 1.979e-09

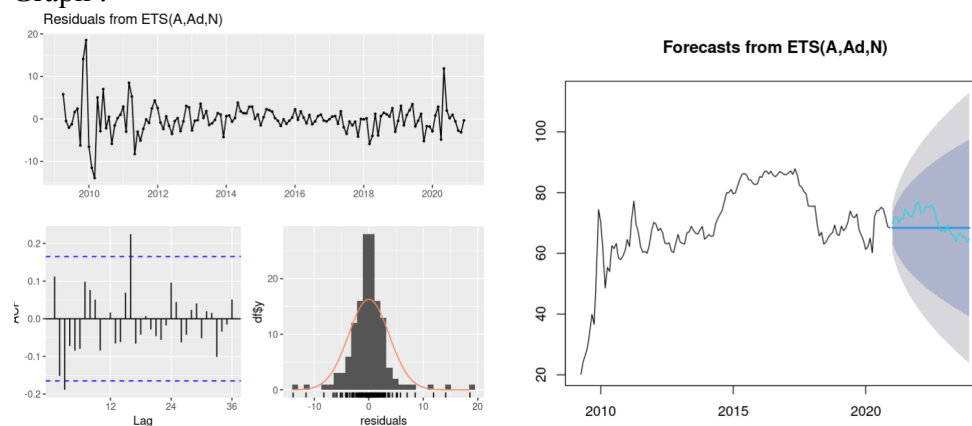
>
> ## Jarque bera test
> ets_jbt <- jarque.bera.test(residuals_extract)
> ets_jbt

Jarque Bera Test

data: residuals_extract
X-squared = 272.01, df = 2, p-value < 2.2e-16

> # Calculate AIC and BIC
> aic_ets <- AIC(ets_fit)
> aic_ets
[1] 1080.228
> bic_ets <- BIC(ets_fit)
> bic_ets
[1] 1097.92
```

Graph :



TBATS Model

```
##-----  
## TBATS model  
# Fit the TBATS model  
tbats_fit <- tbats(train)  
summary(tbats_fit)  
checkresiduals(tbats_fit)  
  
# Forecast future values  
fr <- forecast(tbats_fit, h = length(test))  
plot(fr)  
lines(test, col="turquoise2")  
accuracy(fr, test)  
  
residuals_extract <- residuals(tbats_fit)  
  
## Shapiro Test  
tbats_sw <- shapiro.test(residuals_extract)  
tbats_sw  
  
## Jarque bera test  
tbats_jbt <- jarque.bera.test(residuals_extract)  
tbats_jbt  
  
# Calculate AIC and BIC  
aic_tbats <- tbats_fit$AIC  
aic_tbats  
bic_tbats <- tbats_fit$BIC  
bic_tbats  
  
##-----
```

Console :

```
> ## TBATS model  
> # Fit the TBATS model  
> tbats_fit <- tbats(train)  
> summary(tbats_fit)  
      Length Class  Mode  
lambda          0 -none- NULL  
alpha           1 -none- numeric  
beta            1 -none- numeric  
damping.parameter 1 -none- numeric  
gamma.values     0 -none- NULL  
ar.coefficients  0 -none- NULL  
ma.coefficients  0 -none- NULL  
likelihood       1 -none- numeric  
optim.return.code 1 -none- numeric  
variance         1 -none- numeric  
AIC              1 -none- numeric  
parameters       2 -none- list  
seed.states      2 -none- numeric  
fitted.values    141 ts      numeric  
errors           141 ts      numeric  
x                282 -none- numeric  
seasonal.periods  0 -none- NULL  
y                141 ts      numeric  
call             2 -none- call  
series           1 -none- character  
method           1 -none- character  
> checkresiduals(tbats_fit)  
  
      Ljung-Box test  
  
data:  Residuals from TBATS  
Q* = 21.811, df = 24, p-value = 0.5906  
  
Model df: 0.    Total lags used: 24
```



```

> # Forecast future values
> fr <- forecast(tbats_fit, h = length(test))
> plot(fr)
> lines(test, col="turquoise2")
> accuracy(fr, test)
      ME      RMSE      MAE      MPE      MAPE      MASE
Training set -0.05309376 3.609572 2.377769 -0.5951978 3.942597 0.3142233
Test set     -0.88281332 4.506730 3.931104 -1.6081037 5.705149 0.5194972
      ACF1 Theil's U
Training set -0.007439517 NA
Test set      0.896098173 2.821597

> ## Shapiro Test
> tbats_sw <- shapiro.test(residuals_extract)
> tbats_sw

      Shapiro-Wilk normality test

data:  residuals_extract
W = 0.88709, p-value = 5.979e-09

>
> ## Jarque bera test
> tbats_jbt <-jarque.bera.test(residuals_extract)
> tbats_jbt

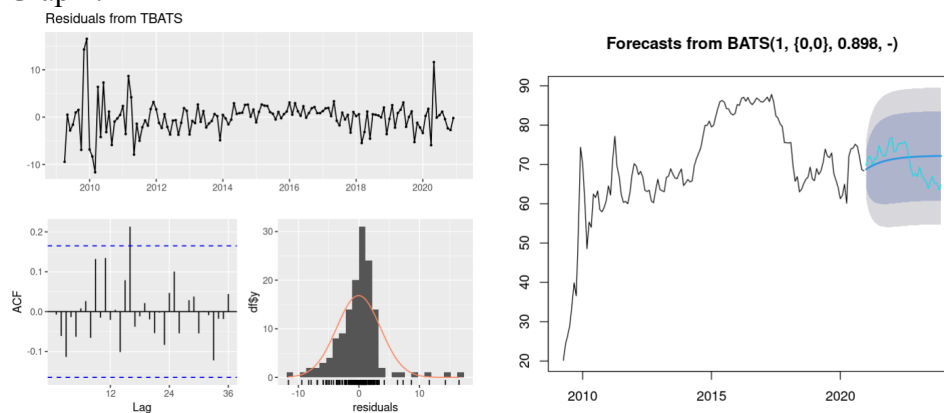
      Jarque Bera Test

data:  residuals_extract
X-squared = 169.62, df = 2, p-value < 2.2e-16

>
> # Calculate AIC and BIC
> aic_tbats <- tbats_fit$AIC
> aic_tbats
[1] 1069.747
> bic_tbats <- tbats_fit$BIC
> bic_tbats
NULL

```

Graph :



Forecast Using the Best Fit Model - Arima (0,1,1)

Console :

```

arima(x = train, order = c(0, 1, 1))

Coefficients:
      ma1
      0.1940
s.e.  0.0838

sigma^2 estimated as 14.18: log likelihood = -384.28, aic = 772.57

Training set error measures:
      ME      RMSE      MAE      MPE      MAPE      MASE      ACF1
Training set 0.2864691 3.751855 2.376287 0.5501767 3.821807 0.9851062 -0.01237035

```

The general form of Arima(p,q,r) is

$$(1-B)Y_t = (1 - \theta_1 B)\varepsilon_t$$

The equation for best fit model of Arima(0,1,1) is

$$Y_t = Y_{t-1} + \varepsilon_t - 0.1940\varepsilon_{t-1}$$

$$Y_t - B(Y_t) = (1 - 0.1940B)\varepsilon_t$$

$$Y_t - B(Y_t) = \varepsilon_t - 0.1940B\varepsilon_t$$

$$Y_t = BY_t + \varepsilon_t - 0.1940B\varepsilon_t$$

$$Y_t = Y_{t-1} + \varepsilon_t - 0.1940\varepsilon_{t-1}$$

Console : Detail Estimation by Arima(0,1,1)

```
> fr
```

	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
Jan 2021	68.4248	63.59946	73.25014	61.04508	75.80452
Feb 2021	68.4248	60.90974	75.93986	56.93151	79.91809
Mar 2021	68.4248	58.95546	77.89414	53.94269	82.90691
Apr 2021	68.4248	57.34055	79.50905	51.47290	85.37670
May 2021	68.4248	55.93269	80.91692	49.31976	87.52984
Jun 2021	68.4248	54.66816	82.18144	47.38583	89.46377
Jul 2021	68.4248	53.51046	83.33914	45.61529	91.23431
Aug 2021	68.4248	52.43638	84.41322	43.97262	92.87699
Sep 2021	68.4248	51.43004	85.41956	42.43355	94.41605
Oct 2021	68.4248	50.48005	86.36956	40.98067	95.86894
Nov 2021	68.4248	49.57788	87.27172	39.60092	97.24868
Dec 2021	68.4248	48.71697	88.13264	38.28427	98.56533
Jan 2022	68.4248	47.89212	88.95748	37.02277	99.82683
Feb 2022	68.4248	47.09915	89.75045	35.81004	101.03957
Mar 2022	68.4248	46.33463	90.51497	34.64080	102.20880
Apr 2022	68.4248	45.59570	91.25390	33.51071	103.33890
May 2022	68.4248	44.87995	91.96965	32.41606	104.43354
Jun 2022	68.4248	44.18532	92.66428	31.35372	105.49588
Jul 2022	68.4248	43.51005	93.33955	30.32098	106.52862
Aug 2022	68.4248	42.85261	93.99699	29.31551	107.53409
Sep 2022	68.4248	42.21165	94.63795	28.33525	108.51435
Oct 2022	68.4248	41.58600	95.26360	27.37839	109.47121
Nov 2022	68.4248	40.97460	95.87500	26.44334	110.40626
Dec 2022	68.4248	40.37652	96.47308	25.52866	111.32094
Jan 2023	68.4248	39.79094	97.05866	24.63308	112.21652
Feb 2023	68.4248	39.21709	97.63251	23.75546	113.09414
Mar 2023	68.4248	38.65430	98.19530	22.89475	113.95485
Apr 2023	68.4248	38.10196	98.74765	22.05001	114.79959
May 2023	68.4248	37.55949	99.29011	21.22039	115.62922
Jun 2023	68.4248	37.02640	99.82320	20.40509	116.44451
Jul 2023	68.4248	36.50221	100.34739	19.60341	117.24619
Aug 2023	68.4248	35.98649	100.86311	18.81469	118.03492
Sep 2023	68.4248	35.47884	101.37076	18.03830	118.81130
Oct 2023	68.4248	34.97890	101.87071	17.27370	119.57590
Nov 2023	68.4248	34.48632	102.36329	16.52037	120.32923
Dec 2023	68.4248	34.00078	102.84882	15.77781	121.07179

Appendix 2

Full Coding

```
# Import dataset
Social_media <- read.csv("social_media.csv")
SocialM <- Social_media[, c(1, 2)]
x <- SocialM[, 1]
y <- SocialM[, 2]
```

```

Data <- ts(y, frequency = 12, start = c(2009, 4), end = c(2023,12))

## plot graph
plot(Data, xlab = "Year ",ylab = "Popularity", main = "Facebook
Popularity between 2009 to 2023")

##-----
# Use train data from 2000 to 2019 for forecasting
train = window(Data, start=2009, end=c(2020,12));
train
# Use remaining data from 2020 to 2024 to test accuracy
test = window(Data, start=2021, end=c(2023,12)); test
autoplot(train)+autolayer(test)
traindt <- ts(train, frequency = 12, start = c(2009, 4), end =
c(2020,12))

## plot graph
plot(traindt, ylab = "Popularity", main = "Facebook Popularity
between 2009 to 2020")

##-----
## Auto Arima
library(forecast)
arima_fit <- auto.arima(train, ic="aic", trace=TRUE)
summary(arima_fit)
checkresiduals(arima_fit)
fr <- forecast(arima_fit, h=length(test))
layout(matrix(c(1,1)))
plot(fr)
lines(test, col="turquoise2")
accuracy(fr, test)

##-----
library(tseries)

# Autocorrelation Function
acf(train, main = "Autocorrelation Function")

# Partial Autocorrelation Function
pacf(train, main = " Partial Autocorrelation Function")

adf.test(train) # model is non-stationary
kpss.test(train) # model is non-stationary

##-----
## Decompostion using Additive - it is a seasonal model
decomposed_comp <- decompose(train, type="additive")
trend <- decomposed_comp$trend
seasonal <- decomposed_comp$seasonal
remainder <- decomposed_comp$random
plot(decomposed_comp)
checkresiduals(remainder)
cbind(decomposed_comp$x,decomposed_comp$trend,decomposed_comp$seaso
nal,decomposed_comp$random)

```

```

##-----
# Define response variable and time variable for the linear model
Y_linear <- train
X_train <- seq(from = 2009, to = 2020, length.out = length(train))
print(X_train)

# Plot the training data with linear model
plot(X_train, Y_linear, col = "blue", xlab = "Year", ylab =
"Popularity", main = "Linear Model of Facebook Popularity")
trend <- lm(Y_linear ~ X_train)

# Add the regression line to the plot
abline(trend, col = "red")

# Summary of the linear model
summary(trend) # significant positive trend

# Check residuals of the linear model
checkresiduals(trend)

##-----

# Calculate the moving average with a window size of 12
movavg <- ma(Data, order = 12)

# Plot the original time series
plot(Data,
      main = "Time Series with Moving Average",
      ylab = "Amount in RM mil",
      xlab = "Year")

# Add the moving average line to the plot
lines(movavg, col = "red", lwd = 2)

##-----
first_diff <- ts(diff(train, lag=12))
ts.plot(first_diff, gpars=list(main= "Seasonal Differences",
                               xlab="Year",
                               ylab="Popularity", lty=1))
acf(first_diff, main="Correlogram of Seasonal Differences")
pacf(first_diff, main="Partial Autocorrelation Function of Seasonal
Differences")

adf.test(first_diff)
pp.test(first_diff)

second_diff <- ts(diff(train, lag=1))
ts.plot(second_diff, gpars=list(main= "Nonseasonal Differences",
                               xlab="Year",
                               ylab="Popularity", lty=1))

acf(second_diff, main="Correlogram of Nonseasonal Differences")
pacf(second_diff, main="Partial Autocorrelation Function of
Nonseasonal Differences")

```

```

adf.test(second_diff)
pp.test(second_diff)

##-----
## Arima

arimafit <- arima(train, order=c(0,1,1));
summary(arimafit)
checkresiduals(arimafit)
fr <- forecast(arimafit, h=length(test))
fr
layout(matrix(c(1,1)))
plot(fr)
lines(test, col="turquoise2")
accuracy(fr, test)

residuals_extract <- residuals(arimafit)

## Shapiro Test
arima_sw <- shapiro.test(residuals_extract)
arima_sw

## Jarque bera test
arima_jbt <-jarque.bera.test(residuals_extract)
arima_jbt

# Calculate AIC and BIC
aic_arima <- AIC(arimafit)
aic_arima

bic_arima <- BIC(arimafit)
bic_arima

##-----
#ETS model
ets_fit <- ets(train)
summary(ets_fit)
checkresiduals(ets_fit)

## Forecasting
fr <- forecast(ets_fit, h=length(test))
plot(fr)
lines(test, col="turquoise2")
accuracy(fr, test)

residuals_extract <- residuals(ets_fit)

## Shapiro Test
ets_sw <- shapiro.test(residuals_extract)
ets_sw

## Jarque bera test
ets_jbt <-jarque.bera.test(residuals_extract)
ets_jbt

```

```

# Calculate AIC and BIC
aic_ets <- AIC(ets_fit)
aic_ets
bic_ets <-BIC(ets_fit)
bic_ets

##-----
## TBATS model
# Fit the TBATS model
tbats_fit <- tbats(train)
summary(tbats_fit)
checkresiduals(tbats_fit)

# Forecast future values
fr <- forecast(tbats_fit, h = length(test))
plot(fr)
lines(test, col="turquoise2")
accuracy(fr, test)

residuals_extract <- residuals(tbats_fit)

## Shapiro Test
tbats_sw <- shapiro.test(residuals_extract)
tbats_sw

## Jarque bera test
tbats_jbt <-jarque.bera.test(residuals_extract)
tbats_jbt

# Calculate AIC and BIC
aic_tbats <- tbats_fit$AIC
aic_tbats
bic_tbats <- tbats_fit$BIC
bic_tbats

##-----

```